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CIS Ethics Approval System

You are Mohammad Danyal (CS2022 - 202318141)

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Application ID: 3188

Title of research:

LinTeX: Promoting Accessibility in LaTeX Documents

Summary of research (short overview of the background and aims of this study):

LinTeX is a new website being developed to combat accessibility problems in LaTeX, with a focus on screen readers. This study aims to understand its effectiveness at targeting this problem and highlighting solutions to its users. Additionally, it aims to identify areas for improvement.

How will participants be recruited?

Participants will be recruited via email and the 4th Year CompSci Strath 2025/26 Discord server in a non-invasive manner. The following message will be used in both contexts:

"Dear <Participant>,"

I would like your views on a tool I'm making that helps people write better LaTeX documents. In particular, it helps authors make documents that meet accessibility needs, and it helps authors with accessibility needs write their LaTeX documents.

LinTeX is a tool being developed to combat this dilemma, placing emphasis on accessibility for screen readers. This user study will determine its effectiveness in doing so.

This user study will involve an in-person think-aloud session taking place on the University of Strathclyde's campus. Further details can be found in the Participant Information Sheet (PIS) attached.

To participate in the user study, you should be a member of the University of Strathclyde, or another higher education facility. Some prior experience with LaTeX will be expected, and participants should be 18-years old or above.

Participation is entirely voluntary and will not affect you in any courses or how you are treated. If you could spare around 30 minutes to participate in the user study, it would be greatly appreciated.

If you are interested in participating in the user study, please contact me via email.

My contact details:

Mohammad Danyal
mohammad.danyal.2023@uni.strath.ac.uk

My supervisor's details:

Brian Mitchell
brian.mitchell@strath.ac.uk

Kind regards,

Mohammad Danyal"

- '<Participant>' will be replaced with the group/individual being contacted.

What will the participants be told about the proposed research study? Either upload or include a copy of the briefing notes issued to participants. In particular this should include details of yourself, the context of the study and an overview of the data that you plan to collect, your supervisor, and contact details for the Departmental Ethics Committee.

PDF File: [View document](#)

Details listed on Participant Information Sheet (PIS).

How will consent be demonstrated? Either upload or include here a copy of the consent form/instructions issued to participants. It is particularly important that you make the rights of the participants to freely withdraw from the study at any point (if they begin to feel stressed for example), nor feel under any pressure or obligation to complete the study, answer any particular question, or undertake any particular task. Their rights regarding associated data collected should also be made explicit.

PDF File: [View document](#)

Details listed on Consent Form.

What will participants be expected to do? Either upload or include a copy of the instructions issued to participants along with a copy of or link to the survey, interview script or task description you intend to carry out.

PDF File: [View document](#)

PDF File: None.

Details listed on the User Study Script.

What data will be collected and how will it be captured and stored? In particular indicate how adherence to the Data Protection Act and the General Data Protection Regulation (GDPR) will be guaranteed and how participant confidentiality will be handled.

Several pieces of data will be stored:

1. Assigning participants unique IDs. This section will have information such as their name and email address (to contact the participant for arranging a time/date). Participant IDs will be created when the participant has requested to join the user study.
2. Extracting details from the user study (such as answers to questions, audible points made by the participant, or unspoken points of note such as how long a specific task has taken to complete) and assigning it to the relevant participants' ID, consequently ensuring related data can be deleted as requested. All details for this group will be acquired from the user study.
3. Anonymising the data from the user study to be used in the final report. This section will have no participant IDs and will have no relation to any participant of the user study. Data will be anonymised at the earliest practicable stage.

Each group of data, as numbered above, will be stored only in separate folders on the university's OneDrive, within password-protected folders, ensuring a folder for each numbered section.

How will the data be processed? (e.g. analysed, reported, visualised, integrated with other data, etc.) Please pay particular attention to describing how personal or sensitive data will be handled and how GDPR regulations will be met.

Anonymised data will be inserted into the final project report after being thoroughly analysed.

Some examples may include:

- How often a listening, or a section of the listening, has been repeated at a stage;
- How often and where a section of the listening has been paused;
- Comparing answers with questions from the initial hearing to the hearing after having used the LinTeX tool;
- Analysing how well the overall document was understood initially to determine difficulties with screen readers;
- Analysing correlations between a user's experience with LaTeX and screen readers with the results from the other components of the user study;
- Mathematical analysis of Likert scale data including frequencies, median, percentages, and mode;
- Comparison of Likert scale data from before and after the think-aloud session.

Results will be presented to minimise the chance of any participant being identifiable. Additionally, any qualitative results will be anonymised.

How and when will data be disposed of? Either upload a copy of your data management plan or describe how data will be disposed.

PDF File: None.

Physical notes taken during the user study will be digitalised as soon as reasonably possible. Once digitalised, they will be securely destroyed.

Electronic data will be securely destroyed on 1st August 2026.

Supervisor's name (students need to complete/staff should probably put N/A).

Brian Mitchell

☒ I confirm that my supervisor has seen and approved both my planned study and this associated ethics application.